

A PROJECT REPORT ON

**AI-Driven Web Platform for Automated Legal
Document Generation Using OCR and NLP
Techniques**

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ABSTRACT

In the finance sector, the verification and documentation process for Realty is both time-consuming and labor-intensive, involving the manual entry of various property documents into spreadsheets and subsequent macro-driven document generation. This project aims to streamline and automate the legal document generation process through the development of a web application. Leveraging the Django Rest Framework (DRF) for backend services, React.js for an intuitive and responsive user interface, and PostgreSQL for robust data management, this application will significantly enhance operational efficiency.

Key functionalities include the automated extraction and validation of critical data such as home and land addresses from scanned documents, which are then used to generate legally compliant documents ready for signing. Additionally, the integration of Artificial Intelligence and Machine Learning components will enable intelligent data verification and error detection, ensuring higher accuracy and reliability. By transitioning from a manual to an automated system, this solution will reduce processing times, minimize human error, and improve overall productivity within the legal verification workflow in banks.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Legal documentation is critical in real estate and finance, particularly for property verification during loans, mortgages, and due diligence. Traditionally, this process involves manual examination of various scanned documents—like land records, sale deeds, and tax receipts—to extract relevant details for report generation. This method is time-consuming, error-prone, and inefficient at scale. The project **Automated Legal Report Generation** offers a web-based solution to streamline this workflow using:

- **OCR:** Extracts text from scanned documents in English and regional languages.
- **NLP and ML:** Parses extracted data, identifies key attributes (e.g., owner name, plot number), and corrects inconsistencies.
- **Django REST Framework:** Handles backend logic and secure API interactions.
- **React.js:** Powers an intuitive frontend for document upload, data review, and report downloads.
- **PostgreSQL:** Stores structured legal data, client information, and report history.

The platform reduces manual effort, ensures legal accuracy, and enables faster, more scalable legal documentation.

1.2 Motivation

The legal documentation process for real estate remains highly manual and inefficient. Professionals must extract and verify data from scanned documents, then generate reports manually—leading to delays and errors.

Automation is necessary to:

- **Reduce Turnaround Time:** Automate extraction and report creation to process more cases faster.
- **Enhance Accuracy:** Reduce human error using ML and LLMs.
- **Improve Scalability:** Handle high volumes without increasing manpower.

- **Ensure Standardization:** Generate consistent, audit-ready reports.
- **Support Multilingual Inputs:** Seamlessly process documents in Marathi and other regional languages.
- **Empower Legal Teams:** Free up professionals for higher-level tasks.

1.3 Problem Definition and Objectives

This project addresses these challenges with the following goals:

- **Automated Data Extraction:** Use OCR + NLP to extract key fields like land area, village, and ownership.
- **AI-Based Validation:** Use LLMs to detect and correct OCR errors via learned patterns.
- **Legal Report Generation:** Generate compliant reports using verified data and dynamic templates.
- **User Interface:** Enable easy upload, review, and download through a modern web app.

1.4 Project Scope and Limitations

Scope

The system automates legal report generation from scanned property documents with:

- **Multilingual OCR Support:** Processes documents like 7/12 Utara in regional languages.
- **Data Parsing and Structuring:** Extracts and formats relevant details using NLP.
- **Template-Based Output:** Inserts validated data into legal templates for final reports.
- **Role-Based Access:** Provides admin tools for reviewing and approving reports.

Limitations

- **OCR Quality Dependency:** Poor scans may lead to inaccurate extraction.
- **Language Variability:** Handwriting styles and layouts can affect model performance.
- **No Real-Time Verification:** System doesn't cross-check data with official government sources.
- **Template Limitations:** Fixed templates may not fit every case without manual customization.

1.5 Methodology

The solution uses a modular pipeline combining OCR, NLP, and web technologies:

- **OCR (Tesseract):** Extracts raw text from scanned images.
- **Text Preprocessing:** Applies regex, spell correction, and tokenization.
- **LLM-Based Parsing:** Converts unstructured text into structured JSON using language models.
- **Web Framework:** Django (DRF) for backend APIs, React.js for frontend UI.
- **Report Generation:** Auto-fills legal templates with verified data to produce downloadable reports.

This architecture ensures the solution is intelligent, user-friendly, and scalable.

CHAPTER 2

LITERATURE SURVEY

2.1 Optical Character Recognition (OCR)

OCR is crucial for converting printed or handwritten documents into machine-readable formats. Traditional OCR systems, relying on template matching and feature extraction, faced challenges in accuracy for regional scripts. The Tesseract engine, as discussed in Patel et al. [2], improved OCR capabilities with support for Unicode and custom language training, making it effective for Indian scripts like Marathi and Hindi. Deep learning models, like Google Vision OCR and PaddleOCR, have advanced OCR technology by improving recognition in noisy documents and complex layouts [6].

2.2 Natural Language Processing (NLP) and Information Extraction

NLP plays a key role in parsing OCR-extracted text. While rule-based systems were once used for entity recognition, the emergence of transformer-based models like BERT and multilingual variants (mBERT, IndicBERT) has greatly enhanced named entity recognition (NER) and data classification in multilingual legal documents [7].

2.3 Document Parsing and Automation in Legal Workflows

The push to automate legal workflows in finance and real estate has spurred research into document parsing and automation. Singh [4] explores OCR-based techniques for parsing forms, while Rao et al. [5] present algorithms for processing tabular data in property documents.

2.4 Multilingual and Domain-Specific Challenges

OCR systems trained on Latin alphabets often struggle with Indian languages due to script complexity and varied fonts [3]. Indian legal documents are typically bilingual, requiring hybrid processing. Additionally, the lack of public datasets for regional documents like 7/12 Utara complicates supervised learning.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

- The users of the system have access to a stable internet connection to interact with the web-based platform.
- The scanned documents are either in image format (JPEG/PNG) or PDF.
- The backend server runs on a Linux-based environment with Python 3.9+ and Django installed. PostgreSQL is used as the relational database system for storing all extracted and processed data.
- The OCR engine (Tesseract) is installed and configured with support for Marathi, Hindi, and English languages.
- External APIs such as Google Cloud Vision (optional) or LLMs such as OpenAI/GPT are assumed to be accessible if required.
- Users are authenticated via role-based login (Admin/Client), and their access rights are enforced throughout the system.

3.2 Functional Requirements

The system features the following core modules and functionalities.

3.2.1 OCR-Based Document Upload and Text Extraction

- **Description:** Upload scanned documents for OCR-based text extraction.
- **Inputs:** Scanned image or PDF.
- **Processing:** Tesseract OCR with language support (Marathi, Hindi, English).
- **Outputs:** Raw extracted text.
- **Actors:** Admin, Client.

3.2.2 Multilingual Text Parsing and Data Structuring

- **Description:** Parses extracted text to identify fields like Village, Plot Number, Owner Name, etc.
- **Inputs:** OCR-extracted text.
- **Processing:** Text classification and segmentation using NLP models.
- **Outputs:** Structured JSON or tabular data.
- **Actors:** Backend Processor.

3.2.3 Document Generation

- **Description:** Automatically generates a legal document from structured data.
- **Inputs:** Structured data.
- **Processing:** Merges data with template using Jinja2 or docxtpl.
- **Outputs:** Downloadable PDF.
- **Actors:** Admin, Client.

3.2.4 Admin Panel for Review and Approval

- **Description:** Admins can review, edit, and approve generated documents.
- **Inputs:** Admin decisions and edits.
- **Processing:** Saves edits and finalizes document.
- **Outputs:** Approved legal report.
- **Actors:** Admin.

3.2.5 User Authentication and Role-Based Access Control

- **Description:** Secure login for Admin and Client with role-specific access.
- **Inputs:** Username and password.
- **Processing:** Authentication with Django's system.
- **Outputs:** Role-specific dashboard.
- **Actors:** Admin, Client.

3.2.6 Data Validation and Error Handling

- **Description:** AI-based validation for data accuracy (e.g., address, numeric values).
- **Inputs:** Structured data.
- **Processing:** Rule-based and model-based error checks.
- **Outputs:** Error logs, warnings, or suggestions.
- **Actors:** Backend Processor.

3.3 External Interface Requirements

3.3.1 User Interfaces

- **Login/Signup Interface:** A secure login/signup page for Admins and Clients with password protection.
- **Dashboard:** Displays a summary of uploaded documents, processing status, generated reports, and system notifications.
- **Upload Interface:** Allows users to upload scanned legal documents (PDF or image formats).
- **OCR Output Review:** Admins can view, validate, and edit the extracted data before generating final reports.
- **Document Viewer:** Displays a preview of the legal document generated before final download.

- **Download Section:** Enables users to securely download their finalized legal reports in PDF format.
- **Error Handling and Notifications:** Users are informed of failed uploads, incorrect formats, or OCR errors through user-friendly alerts.

3.3.2 Hardware Interfaces

- **Client Side:**
 - CPU: Dual-core or higher
 - RAM: 4 GB or higher
 - Storage: 100 MB for browser cache/data
- **Server Side:**
 - CPU: Quad-core or higher
 - RAM: 8 GB or higher
 - Storage: Minimum 50 GB for scanned documents, logs, and generated reports
 - GPU (Optional): For model inference in OCR/LLM if required

3.3.3 Software Interfaces

1. **Operating System:** Compatible with macOS, Linux, or Windows for both server and client.
2. **Web Browser:** Chrome, Firefox, Safari, or Edge (latest versions recommended).
3. **Backend Framework:** Django Rest Framework (Python-based).
4. **Frontend Framework:** React.js for dynamic UI.
5. **Database:** PostgreSQL for storing structured data.
6. **OCR Engine:** Tesseract OCR with multilingual language packs (Marathi, Hindi, English).
7. **LLM APIs (Optional):** Integration with OpenAI/GPT for intelligent text parsing and error correction.
8. **Python Libraries:** Pandas, NLTK, SpaCy, PyMuPDF, Pillow, and other supporting libraries.

3.3.4 Communication Interfaces

- **HTTP/HTTPS Protocol:** For client-server communication via RESTful APIs.
- **JSON:** Data exchange format between frontend and backend.
- **Email Notifications:** Integration with SMTP for sending user confirmations, approvals, and document readiness alerts.
- **Cloud Storage (Optional):** Integration with AWS S3 or Google Cloud Storage for storing documents securely.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- The system should be able to process and extract data from a scanned document within 10 seconds on average.
- Concurrent processing should be supported for at least 10 documents at a time without performance degradation.
- Web application response time should be less than 2 seconds for standard actions like document upload and report download.

3.4.2 Safety Requirements

- The system must maintain data integrity even in case of sudden shutdowns or crashes.
- All document data should be auto-saved at each processing step to avoid data loss.
- Proper error messages should be shown to the user in case of failure, and logs should be maintained for future debugging.

3.4.3 Security Requirements

- All user data must be stored securely using database encryption mechanisms.
- Sensitive documents must be transmitted only over HTTPS.

- Role-based access control should ensure that only Admins can approve or edit final reports.
- User authentication should include secure password storage (bcrypt/argon2) and optional 2FA.
- Uploaded documents must be scanned for malware and securely deleted after processing if required.

3.4.4 Software Quality Attributes

- **Usability:** The UI must be intuitive and support both technical and non-technical users.
- **Reliability:** The system should ensure consistent results with minimal downtime.
- **Maintainability:** Modular architecture and well-documented code-base to ease future enhancements.
- **Portability:** The application should be deployable across major platforms and hosting environments.
- **Scalability:** Capable of handling increased user loads and document volume over time.

3.5 System Requirements

3.5.1 Database Requirements

- PostgreSQL will be used for data storage.
- Tables will include users, uploaded documents, extracted fields, generated reports, and audit logs.
- Relational integrity will be enforced via foreign keys.
- Backup and restore features should be enabled for data safety.

3.5.2 Software Requirements (Platform Choice)

- **Frontend:** React.js (Node.js for build)
- **Backend:** Django Rest Framework (Python 3.10+)
- **OCR Engine:** Tesseract OCR with language support (English, Hindi, Marathi)
- **Document Generation:** Python-docx, ReportLab, or similar libraries
- **Deployment:** Docker (optional), Heroku, or AWS EC2
- **Others:** GitHub for version control, Postman for API testing

3.5.3 Hardware Requirements

Client Side:

- Modern web browser (Chrome, Firefox, Safari)
- 4GB RAM, Dual-core CPU

Server Side:

- 8GB+ RAM, Quad-core CPU
- Optional GPU for LLM inference
- Minimum 50 GB of storage for documents and logs

3.6 Analysis Models: SDLC Model to be Applied

For this project, the **Iterative Incremental Model** is used.

- **Justification:** Since the requirements are fairly clear but may evolve based on stakeholder feedback, an iterative approach allows for gradual development and continuous integration.
- Each increment includes requirement analysis, design, implementation, and testing phases.
- It allows the core document extraction and generation features to be delivered early, with incremental improvements like LLM integration, language expansion, and user feedback handling added in subsequent phases.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system adopts a client-server architecture, enabling seamless interaction between the user interface, backend services, and OCR/ML engines.

1. Overview of Architecture

The system follows a three-tier architecture:

- **Presentation Layer (Frontend):** Built using `React.js`, this layer provides an intuitive and responsive interface for users. It communicates with the backend via RESTful APIs.
- **Application Layer (Backend/API):** Developed using `Django Rest Framework (DRF)`, this layer handles request processing, document routing, validation logic, business rules, and report generation.
- **Data Layer (Database and Storage):** Uses `PostgreSQL` for structured storage of user data, document metadata, extracted fields, and audit logs.

2. OCR and NLP Integration

To support multilingual data extraction, the system integrates:

- **OCR Engine:** Tesseract OCR with language packs to extract text from scanned PDFs or images.
- **LLM/NLP Module:** A lightweight language model (e.g., `spaCy` or `Azure/GPT APIs`) is used for entity recognition and extraction of legal field values.

3. Workflow Pipeline

1. **Document Upload:** The user uploads scanned legal documents through the frontend.
2. **OCR Processing:** The backend invokes Tesseract to extract raw text from the document.
3. **Information Extraction:** Extracted text is passed through NLP modules to identify and extract relevant fields.

4. **Validation:** Extracted data is validated and corrected using predefined rules or ML-based auto-correction models.
5. **Document Generation:** Once validated, the system fills a pre-designed legal document template with the extracted data.
6. **Admin Review:** The final document is shown to the admin for approval or editing.
7. **Report Download:** After approval, users can download the finalized legal report in PDF format.

4. Security and Access Control

- **Authentication:** Role-based access with secure login credentials. Only admins can approve or edit final documents.
- **Data Protection:** Documents are encrypted during storage and transmission. HTTPS ensures secure communication.
- **Audit Logs:** Every action (upload, modify, approve) is logged for traceability.

5. Deployment

The system can be deployed on:

- **Cloud Services:** AWS EC2 or Heroku for hosting backend and frontend.
- **Docker Containers:** For portability, the services can be containerized.
- **CI/CD Pipelines:** GitHub Actions or Jenkins for automated testing and deployment.

6. Scalability and Extensibility

The architecture is designed to be modular:

- **Plug-and-Play ML Modules:** New models can be easily integrated for better accuracy.
- **Template Flexibility:** New document formats can be introduced by defining corresponding templates.

- **Language Expansion:** Support for new languages can be added by extending the OCR/NLP modules.

7. Diagram

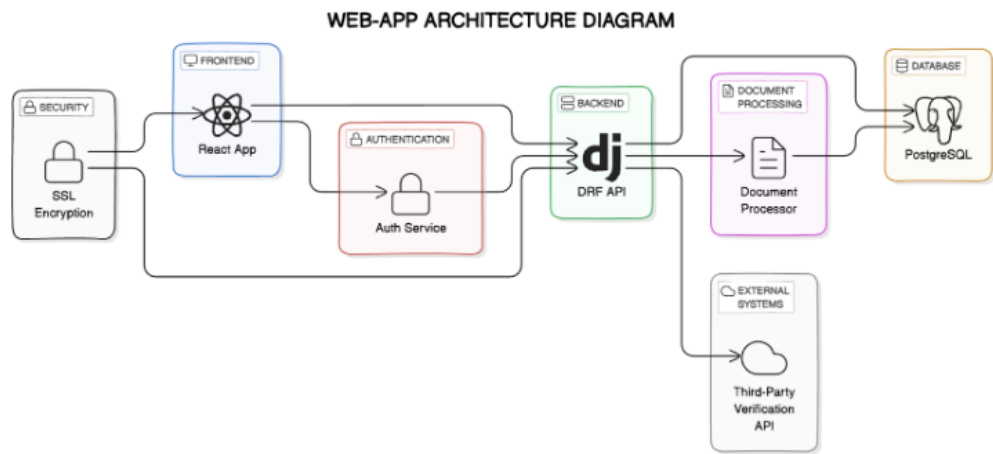


Figure 4.1: System Architecture of Automated Legal Report Generation

4.2 Flow Chart

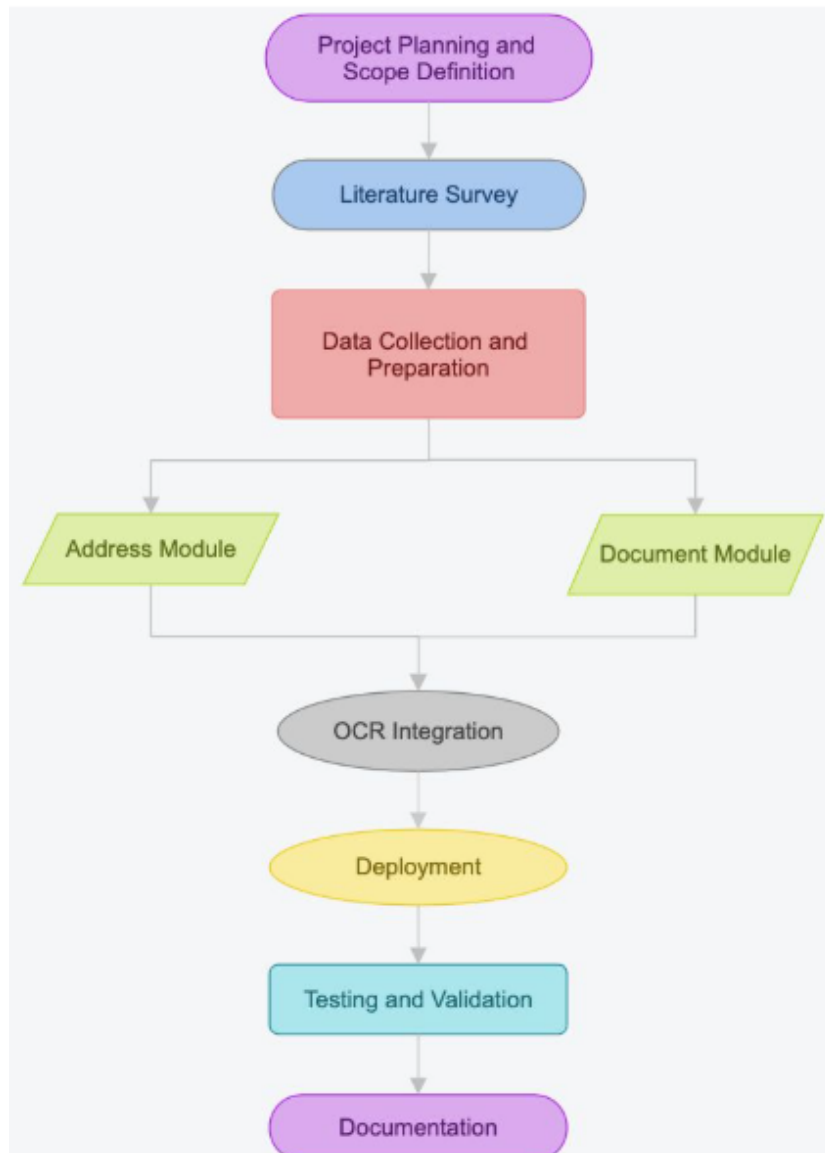


Figure 4.2: System Flow Chart

4.3 High Level Design Diagram

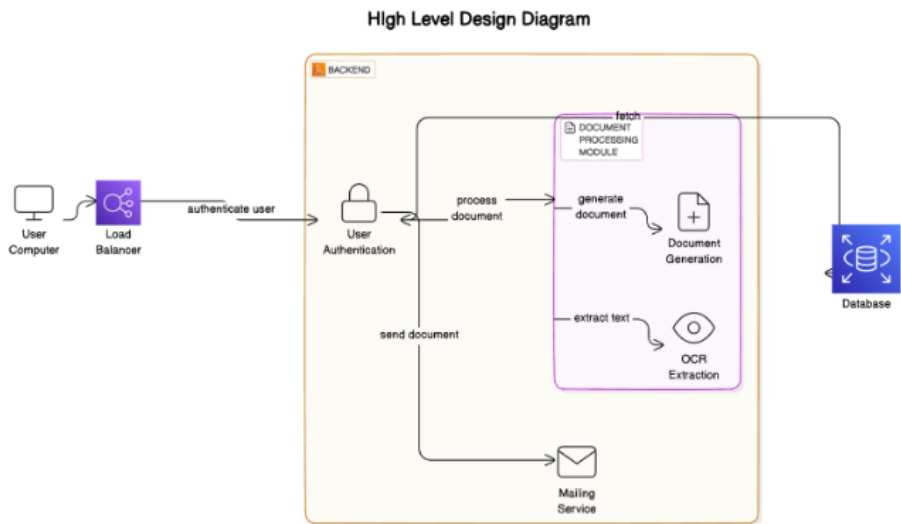


Figure 4.3: High Level Design Diagram

4.4 Class Diagram

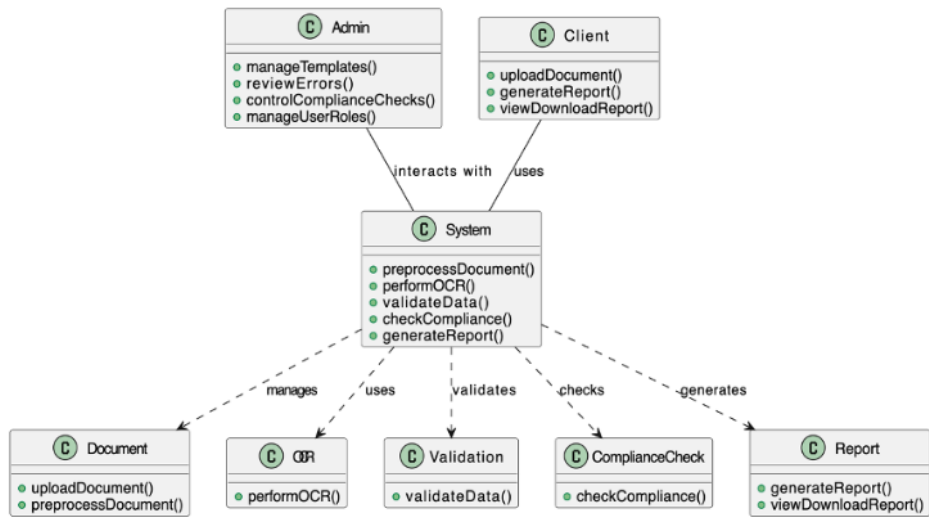


Figure 4.4: Class Diagram of the System

4.5 Use Case Diagram

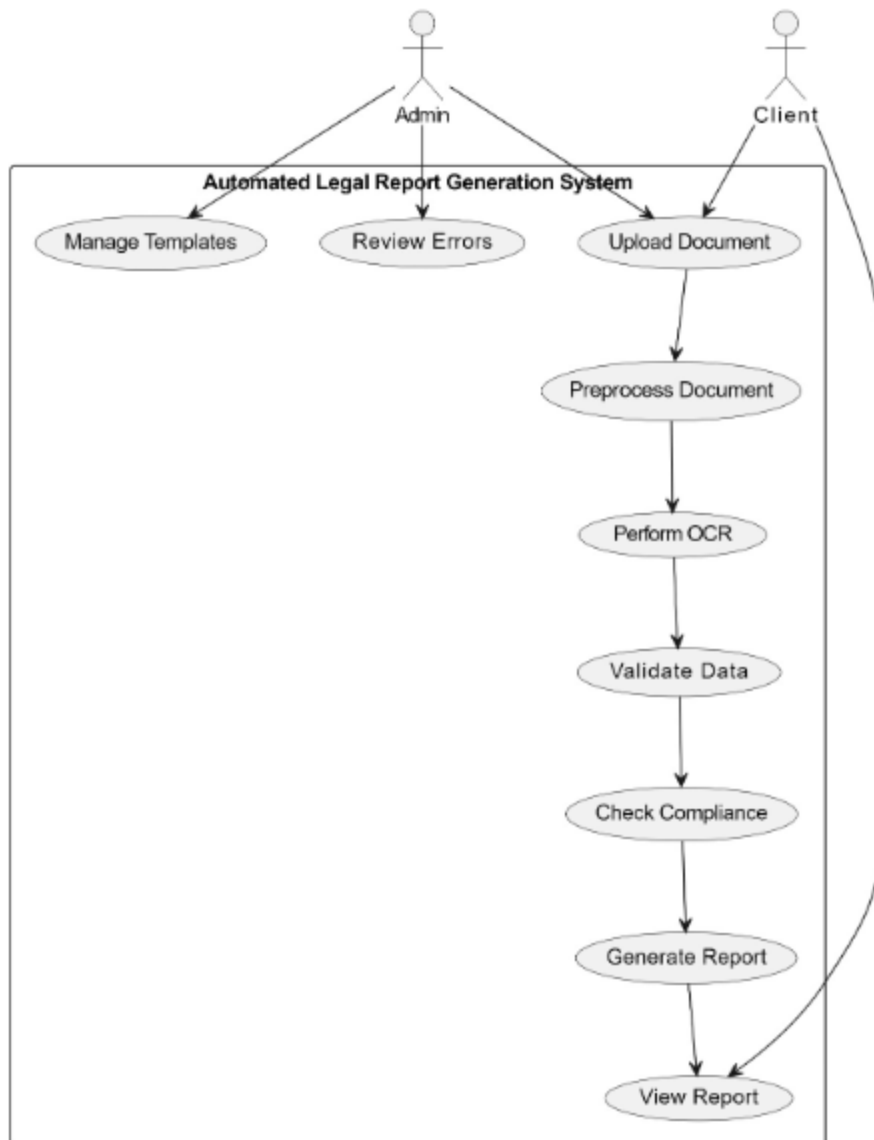


Figure 4.5: Use Case Diagram

4.6 Sequence Diagram

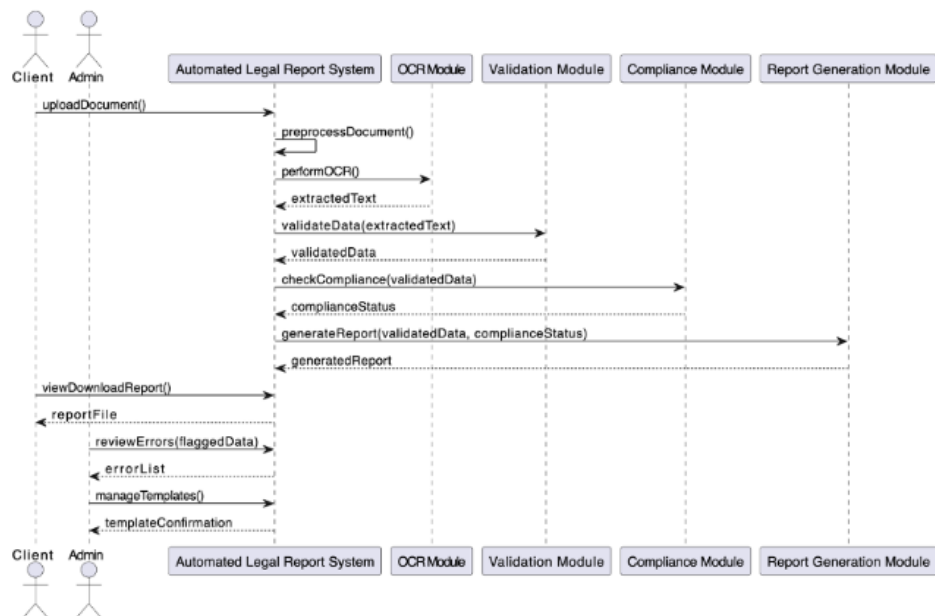


Figure 4.6: Sequence Diagram of the System

4.7 Deployment Diagram

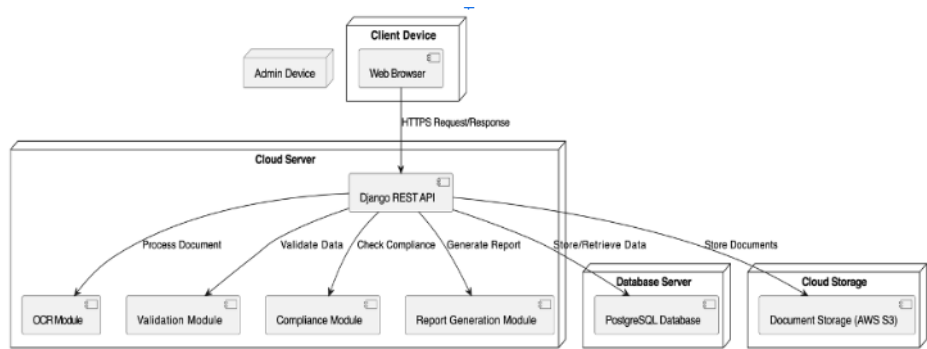


Figure 4.7: Deployment Diagram of the System

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

Reconciled estimates are finalized time and effort projections based on ongoing evaluation during each development phase. These reflect a more accurate understanding of task complexities and resource availability compared to initial rough estimates.

- **Requirement Gathering and Analysis (1 week):** This phase included stakeholder meetings, market research, legal process understanding, and drafting the Software Requirement Specification (SRS).
- **System Design (1 week):** This included high-level architectural planning, component definition, and preparation of various diagrams such as flowcharts, use case diagrams, and class diagrams.
- **Frontend and Backend Development (3–4 weeks):** This was the core phase, covering:
 - UI/UX design using React.js.
 - Backend development using Django REST Framework.
 - Integration of OCR using Tesseract and NLP processing with spaCy.
 - Secure user authentication, form submission, and legal document generation logic.
 - API design, validation, and testing.
- **Database Design and Integration (0.5 week):** Designed using PostgreSQL, integration was done with the Django ORM.
- **Testing and Debugging (1.5 weeks):** Included writing unit tests, performing integration and functional testing, validating OCR/NLP outputs, and debugging performance issues.
- **Deployment and Hosting (0.5 week):** Environment setup, CI/CD configurations, and security settings (HTTPS, firewalls, user roles) were managed in this phase.
- **Documentation and Final Report Writing (1 week):** Comprehensive documentation was prepared including user manuals, code documentation, architecture diagrams, and the final project report in LaTeX.

5.1.2 Project Resources

- **Human Resources:** A team of 4 developers contributed to different segments of the project:
 - **Frontend Developer:** Focused on the React.js interface, styling, and UX.
 - **Backend Developer:** Implemented Django APIs, database models, and authentication.
 - **AI/NLP Developer:** Integrated Tesseract OCR, preprocessed scanned documents, and used spaCy for entity recognition and classification.
 - **DevOps/Tester:** Handled deployment to AWS, created CI/CD pipelines, and performed automated/manual testing.
- **Software Tools:**
 - **Frontend:** React.js, Tailwind CSS, Axios, Postman
 - **Backend:** Django REST Framework, Python 3.10
 - **Database:** PostgreSQL, pgAdmin
 - **AI/OCR:** Tesseract OCR, spaCy, Pandas
 - **Version Control:** Git, GitHub
 - **IDE Editors:** Visual Studio Code, Jupyter Notebook
- **Hardware Resources:**
 - Development machines with at least 8GB RAM, Intel i5/i7 processors.
 - Smartphones and tablets used for testing responsive design and OCR capabilities.
 - AWS Cloud Services:
 - * EC2 for app hosting
 - * S3 for static file storage
 - * IAM for secure role-based access
- **Informational Resources:**
 - Official documentation for Django, React, Tesseract, and AWS.
 - Stack Overflow, GitHub repositories, and developer blogs.
 - Research papers on document digitization, OCR techniques, and Indian legal report formats.

5.2 Risk Management

5.2.1 Risk Identification

Risk identification is the first crucial step in the risk management process. This phase involves systematically identifying potential risks that could affect the project's objectives, timeline, quality, or budget. Below are some of the key risks identified during the project lifecycle:

- **Technology Integration Challenges:** The integration of third-party libraries like Tesseract for OCR and NLP tools into the backend (Django REST framework) could lead to issues with compatibility, performance, or unexpected behaviors.
- **Data Quality Issues:** Poor quality of scanned documents (e.g., low resolution, complex fonts, noisy backgrounds) may reduce the accuracy of the OCR system affecting data processing.
- **Schedule Delays:** Delays could occur potentially impacting the overall timeline.
- **Security Risks:** The project deals with sensitive legal data, and failure to implement strong security measures may expose the system to data breaches, unauthorized access, or tampering.
- **Budget Overruns:** Increased costs due to additional services could potentially exceed the allocated project budget.
- **Performance and Scalability Issues:** The system may face performance challenges, especially when handling large volumes of legal documents potentially affecting system scalability and responsiveness.

5.2.2 Risk Analysis

Risk analysis is the process of evaluating the identified risks in terms of their likelihood and potential impact. This helps to prioritize risks, allowing the team to focus efforts on the most critical risks first. The analysis of the risks identified in the previous section is as follows:

- **Technology Integration Challenges:**
 - **Likelihood:** Medium - Technology integration can be challenging, but with careful selection of libraries and thorough testing, it is manageable.

- **Impact:** High - If the OCR and NLP tools do not integrate smoothly, it could significantly affect the project timeline and functionality, especially in the backend data processing.
- **Data Quality Issues:**
 - **Likelihood:** High - OCR systems often face challenges with low-quality scans, especially with varied fonts or poor image quality.
 - **Impact:** High - Poor data extraction would directly affect the system's output, making legal reports inaccurate, which is critical for the project's success.
- **Security Risks:**
 - **Likelihood:** Low - As long as best practices for security (e.g., HTTPS, user authentication) are followed, this risk is minimal.
 - **Impact:** High - Any security breach would severely affect the trustworthiness of the system, especially when handling sensitive legal documents.
- **Budget Overruns:**
 - **Likelihood:** Low - The budget is closely monitored, and costs are tracked throughout the project lifecycle.
 - **Impact:** Medium - Overruns could reduce profitability or require additional funds, but this can be managed with careful financial tracking and adjustment.
- **Performance and Scalability Issues:**
 - **Likelihood:** Medium - Performance issues may arise when scaling the system, but these can be identified and addressed through stress testing.
 - **Impact:** Medium - Performance issues could hinder user experience, but they can be mitigated with optimization and load balancing techniques.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Risk mitigation, monitoring, and management are ongoing processes that ensure the project can handle risks effectively.

- **Risk Mitigation:** Mitigation strategies are implemented to reduce the likelihood or impact of risks.
- **Risk Monitoring:** Regular monitoring helps identify any emerging risks and track the progress of risk mitigation strategies
- **Risk Management:** Active risk management involves making informed decisions about how to respond when a risk materializes.
- **Continuous Improvement:** Risk management is not a one-time task but an iterative process. Lessons learned from past risks are documented, and improvements are made to the risk management process for future phases of the project.

5.3 Project Schedule

5.3.1 Project Task Set

The tasks for the Automated Legal Report Generation project are divided into several phases:

- **Requirement Gathering and Analysis:** This phase involves creating the project scope and documentation.
- **System Design:** This includes creating wireframes, class diagrams, and other design artifacts.
- **Implementation and Development:** This includes frontend development (React.js), backend development (Django REST Framework), and integration of OCR tools like Tesseract.
- **Testing and Debugging:** Performing unit tests, integration tests, system tests, and user acceptance tests (UAT) to ensure the system works as expected.
- **Deployment:** Deploying the system to a production environment, including configuring cloud services (AWS), setting up continuous integration/continuous deployment (CI/CD), and ensuring that the system is scalable and secure.
- **Documentation and Report Writing:** Preparing the final report, documenting the system architecture, codebase, and user manual. This also includes writing project-related documents and compiling deliverables for stakeholders.

5.3.2 Task Network

In the case of the Automated Legal Report Generation project, the task network would outline the dependencies between phases, such as:

- **Requirement Gathering and Analysis** precedes **System Design**
- **System Design** must be completed before **Implementation and Development**
- **Implementation and Development** is interdependent with **Testing and Debugging**
- **Deployment** is dependent on **Testing and Debugging**
- **Documentation and Report Writing** begins after the majority of the development and testing phases have been completed and data for documentation is available.

The task network can be visualized using a Gantt chart or similar project management tool, which displays task dependencies and critical paths.

5.3.3 Timeline Chart

For the Automated Legal Report Generation project, the timeline chart look like this:

- **Requirement Gathering and Analysis:** 1 week
- **System Design:** 1 week
- **Implementation and Development:** 3–4 weeks
- **Testing and Debugging:** 1.5 weeks
- **Deployment:** 1 week
- **Documentation and Report Writing:** 1 week

The timeline chart can also be used to monitor the progress and adjust the schedule if tasks are delayed or ahead of schedule.

5.4 Team Organization

An organized and effective team is essential for the success of any project. The team structure and communication protocols determine how well team members collaborate and deliver the project's objectives.

5.4.1 Team Structure

Role	Responsibility
Project Manager	Oversees planning, execution, resource allocation, and issue resolution.
Frontend Developer	Designs the UI/UX using React.js to meet user requirements.
Backend Developer	Builds the core logic with Django REST Framework, handling data and API security.
OCR/NLP Expert	Integrates Tesseract and NLP tools for document text extraction and processing.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

6.1.1 System Modules

Module	Description
User Authentication and Authorization	Manages login, logout, and role-based access control for admins and clients.
OCR Integration	Extracts text and property details from scanned documents using Tesseract and NLP.
Report Generation	Converts extracted data into formatted legal reports using templates and outputs PDFs.
Database Management	Uses PostgreSQL and Django ORM to store client info, legal data, and report history.
Frontend Interface	Built with React.js; includes forms, dropdowns, and document upload UI.
Backend API	Developed with Django REST Framework to process requests and connect modules.
Notification and Email System	Sends report emails and alerts via SMTP to keep clients updated.

6.2 Tools and Technologies Used

Technology	Purpose
Django REST Framework	Backend API development for managing reports and documents.
React.js	Builds dynamic and responsive frontend UI.
PostgreSQL	Relational database for storing user and document data.
Tesseract OCR	Extracts text from scanned legal documents.
spaCy	Processes OCR text for extracting key legal information.
AWS EC2 & S3	Hosts application and stores uploaded/generated documents.
Docker	Containerizes application for easy deployment and consistency.
Git & GitHub	Version control and collaborative code management.
Jest & PyTest	Frontend and backend automated testing frameworks.
SMTP	Sends email notifications and reports to clients.

6.3 Algorithm Details

The following algorithms are used to process and generate legal reports based on scanned documents. Each algorithm serves a distinct purpose, ensuring the accuracy and reliability of the extracted information.

6.3.1 Algorithm 1: OCR Text Extraction and Preprocessing

6.3.1.1 OCR and NLP Pipeline

Step	Description
Document Preprocessing	Enhances image quality using resizing, noise removal, and binarization.
Text Extraction	Tesseract OCR converts the preprocessed image into machine-readable text.
Text Cleaning	Removes unwanted characters and symbols using regular expressions.
Tokenization & Segmentation	Splits text into tokens and segments it into structured sections.

6.3.2 Algorithm 2: Named Entity Recognition (NER) for Legal Data Extraction

6.3.2.1 Entity Extraction and Structuring

Step	Description
Entity Recognition	Uses spaCy to identify named entities like names, dates, and addresses.
Custom Legal Rules	Applies patterns to extract legal-specific entities such as case numbers.
Entity Validation	Validates extracted entities using a legal terms list for accuracy.
Data Structuring	Organizes validated entities into a format ready for report generation.

6.3.3 Algorithm 3: Report Generation and Template Filling

6.3.3.1 Report Generation Process

Step	Description
Template Selection	Chooses a report template based on the report type.
Data Mapping	Maps extracted data (names, dates, property info) to template fields.
Text Formatting	Formats text with legal styling rules (alignment, fonts, tables).
PDF Generation	Converts the formatted report into a PDF using ReportLab/WeasyPrint.
Storage and Delivery	Stores the PDF on AWS S3 and sends the client a download link via email.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

7.1.1 Testing Strategy

Testing Type	Description
Unit Testing	Validates individual components using PyTest and Jest.
Integration Testing	Ensures modules work together, e.g., OCR to report generation.
Functional Testing	Tests features like document upload and report generation.
System Testing	Verifies overall system performance and workflow end-to-end.
User Acceptance Testing (UAT)	Confirms system meets real-world user expectations.
Performance Testing	Assesses response time, load handling, and scalability.
Security Testing	Detects vulnerabilities using tools like OWASP ZAP.
Regression Testing	Ensures new changes don't break existing functionality.

7.2 Test Cases & Test Results

7.2.1 Test Cases Summary

Test Case	Description	Input	Expected Output	Result
OCR Text Extraction	Validate OCR on scanned PDFs	Scanned legal document	Accurate text extraction	Pass
NER for Legal Terms	Extract legal entities using NLP	Preprocessed OCR text	Entities like names, dates, addresses	Pass
Report Generation	Generate PDF from extracted data	Legal entities from OCR + NLP	Filled legal report in PDF	Pass
User Auth & Access	Test login and role-based access	Admin and client credentials	Correct access by role	Pass
Email Notification	Confirm report email delivery	Generated legal report	Email with download link	Pass
Performance Under Load	Simulate 1000 users	Concurrent report generation	Stable performance under load	Pass
Security Testing	Test against SQL injection & breaches	Malicious inputs	Block threats, protect access	Pass

CHAPTER 8

RESULTS

8.1 Outcome & Screen Shots

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'psychic-couscous-v99vr44qgrfoxy-3000.app.github.dev/new-entry'. The browser address bar shows the URL. The page content includes a sidebar with a menu icon and a main form area. The form is titled 'Combined Form' and has a subtitle 'Enter all the details'. It contains several input fields and dropdown menus. The fields are arranged in two columns. The first column includes APPID1, BORROWER NAME, LOGIN DATE (with a date picker), DOCUMENT RETURN DATE (with a date picker), REPORT TYPE (a dropdown menu), and HARD COPY SUBMITTED. The second column includes APPID2, WORK SCOPE, LOGIN TYPE, REPORT DATE (with a date picker), OUTCOME (a dropdown menu), and LEGAL ENTITY NAME. Below the form is a section titled 'Client Data' with a field for GROUP NAME (a dropdown menu) and a field for LEGAL ENTITY NAME. The browser's taskbar at the bottom shows the Windows logo, a search bar, and several application icons. The system clock in the bottom right corner shows the time as 12:17 AM on 5/4/2025.

Figure 8.1: Image 1

The screenshot shows a web browser window with a single tab titled 'Create Next App'. The browser address bar shows the URL 'localhost:3000/new-entry'. The page content includes a sidebar with a menu icon and a main form area. The form is titled 'Combined Form' and has a subtitle 'Enter all the details'. It contains several input fields and dropdown menus. The fields are arranged in two columns. The first column includes APPID1, BORROWER NAME, LOGIN DATE (with a date picker), DOCUMENT RETURN DATE (with a date picker), REPORT TYPE (a dropdown menu), and HARD COPY SUBMITTED. The second column includes APPID2, WORK SCOPE, LOGIN TYPE, REPORT DATE (with a date picker), OUTCOME (a dropdown menu), and LEGAL ENTITY NAME. Below the form is a section titled 'Client Data' with a field for GROUP NAME (a dropdown menu) and a field for LEGAL ENTITY NAME. The browser's taskbar at the bottom shows the Windows logo, a search bar, and several application icons. The system clock in the bottom right corner shows the time as 03:14 on 15-05-2025.

Figure 8.2: Image 2

AI-Driven Web Platform for Automated Legal Document Generation Using OCR and NLP Techniques

Building Data

UNIT NO:

UNIT TYPE:

FLOOR:

BUILDING NAME:

BUILDING NO:

PROJECT NAME:

PROJECT PHASE:

CONSTRUCTION LOCAL BODY NAME:

CURRENT LOCAL BODY NAME:

WING:

FLAT AREA:

AREA UNITS:

CHS NAME:

SUBURB LOCALITY:

Figure 8.3: Image 3

OCR Document Processing

Upload a PDF or image file to extract text using OCR technology

Upload Document

Choose File No file chosen

Supported formats: PDF, JPG, PNG, TIFF

Process Document

Figure 8.4: Image 4

Admin Panel Add New

Sr No.	App ID 1	App ID 2	Outcome	Borrower Name	Clerk Name	Login Date	Document Return Date	Last Saved Date	View Client Info	View Building Info	View Land Info	View CHS Info	View Share Certificate	Report Type
7	APP123456	APP654321	Approved	John Doe	Alice Smith	9/26/2024	10/2/2024	9/30/2024	View	View	View	View	View	Valuation Report
8	APP123456	APP654321	Approved	John Doe	Alice Smith	9/26/2024	10/2/2024	9/30/2024	View	View	View	View	View	Valuation Report
9	8ATKDQ1P	MA1350XA		EMQU5JCZ		4/1/2024	12/18/2023	1/1/1970	View	View	View	View	View	
10	0P7SQBXF	YMAFTVSK		EZTO1F6O		11/16/2021	1/17/2023	1/1/1970	View	View	View	View	View	

Figure 8.5: Image 5

AI-Driven Web Platform for Automated Legal Document Generation Using OCR and NLP Techniques

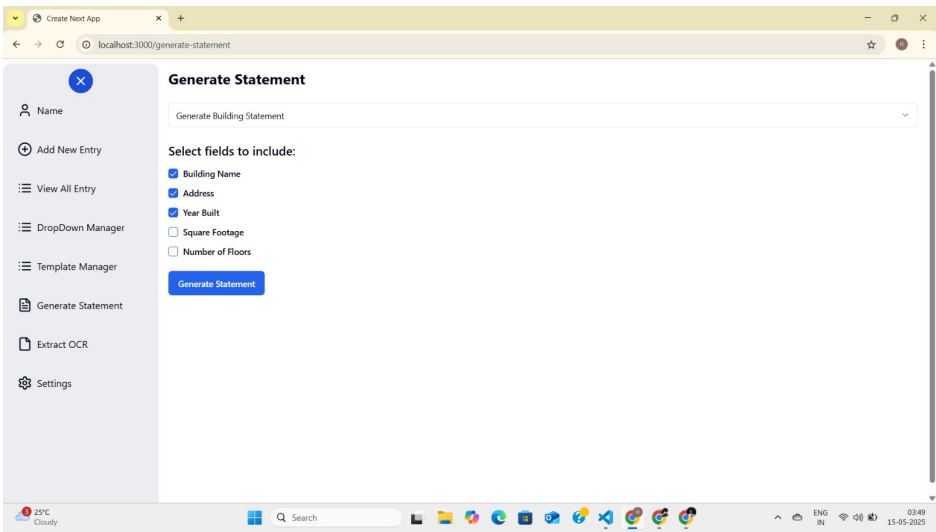


Figure 8.6: Image 6

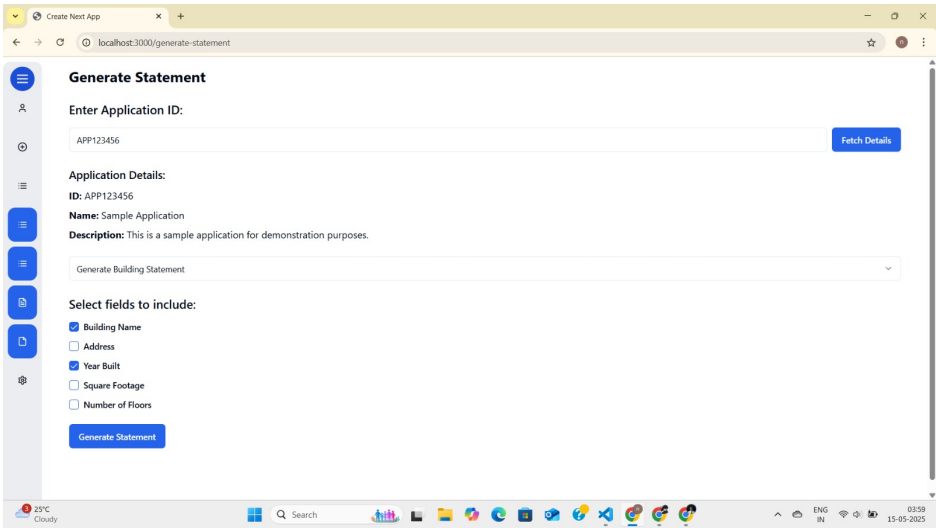


Figure 8.7: Image 7

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

- **Accuracy:** OCR and NLP enabled reliable extraction of legal and property data.
- **Scalability:** Cloud-based setup (AWS) supports high user and document volume.
- **User Experience:** Simple UI with real-time updates and notifications.
- **Security:** Secure data handling with role-based access.

9.2 Future Work

- **Legal Integration:** Connect with external databases for real-time updates.
- **Collaboration Tools:** Enable live edits and comments on documents.
- **Analytics:** Extract trends and risks from legal data.
- **OCR Improvement:** Train custom OCR on legal and handwritten text.
- **Workflow Tools:** Add task tracking and report status updates.
- **Cloud Optimization:** Use Docker/Kubernetes for better scalability.

9.3 Applications

- **Real Estate & Finance:** Automates property and loan documentation.
- **Government:** Generates audit and compliance reports.
- **Corporates:** Handles contracts and governance docs efficiently.
- **Healthcare:** Automates claims and legal medical forms.
- **Education:** Generates admin/legal academic documents.
- **E-Discovery & Notary:** Accelerates reviews and remote notarization.

CHAPTER 10

APPENDIX

10.1 Appendix A: Problem Statement Feasibility Assessment

Satisfiability Analysis

The system must determine whether transformation rules ϕ can convert a scanned document D into a valid report R :

$$\exists x : \phi(x, D) \Rightarrow R$$

If such x exists, the system succeeds; otherwise, fallback mechanisms are required. Since satisfiability is not guaranteed for every document, the problem is non-trivial and partially undecidable.

Computational Complexity

The process involves OCR, NLP, entity extraction, and template matching. These tasks are:

- Verifiable in polynomial time (class ****NP****),
- Not always solvable efficiently due to ambiguity and variability,
- Composed of sub-problems such as:
 - SAT problem – validating logical rules,
 - Subset-Sum – mapping entities to fields,
 - Graph Matching – aligning document layouts to templates.

Thus, the system overall can be considered ****NP-Hard**** in the general case.

Formal Models and Algebraic Structures

Formal systems help in modeling and simplifying system behavior:

- **Finite Automata and CFGs** – for pattern recognition and syntax parsing,
- **Algebraic Structures** – to model transformations on data sets,
- **Turing Machines** – to represent the overall system workflow.

While some components are solvable in deterministic polynomial time, semantic ambiguity often requires heuristics or partial human oversight.

10.2 Appendix B

Survey Paper Details

- **Paper Status:** Accepted
- **Name of the Conference:** International Research Journal Of Modernization in Engineering Technology and Science
- **Paper Title:** Comprehensive Survey on Optical Character Recognition (OCR).



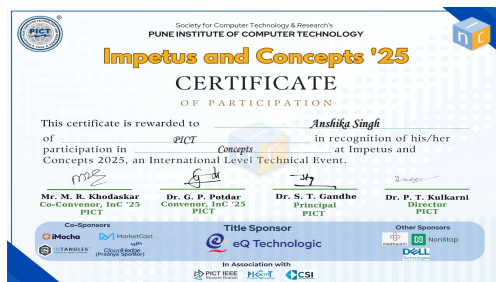
AI-Driven Web Platform for Automated Legal Document Generation Using OCR and NLP Techniques



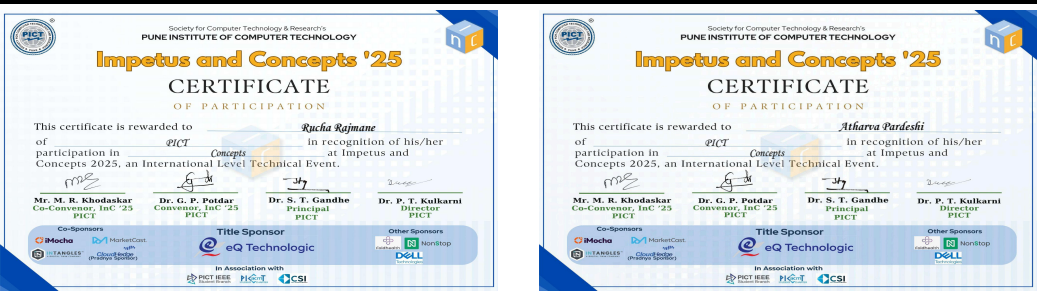
Implementation Paper Details

- **Paper Status:** Under Review
- **Name of the Conference:** NA
- **Paper Title:** Multilingual OCR and LLM-Based Information Extraction for Legal Documents.

Event Participation Details



AI-Driven Web Platform for Automated Legal Document
Generation Using OCR and NLP Techniques



10.3 Appendix C: Plagiarism Report of Project Report

Screenshots of Plagiarism Report

The plagiarism report is attached herewith.

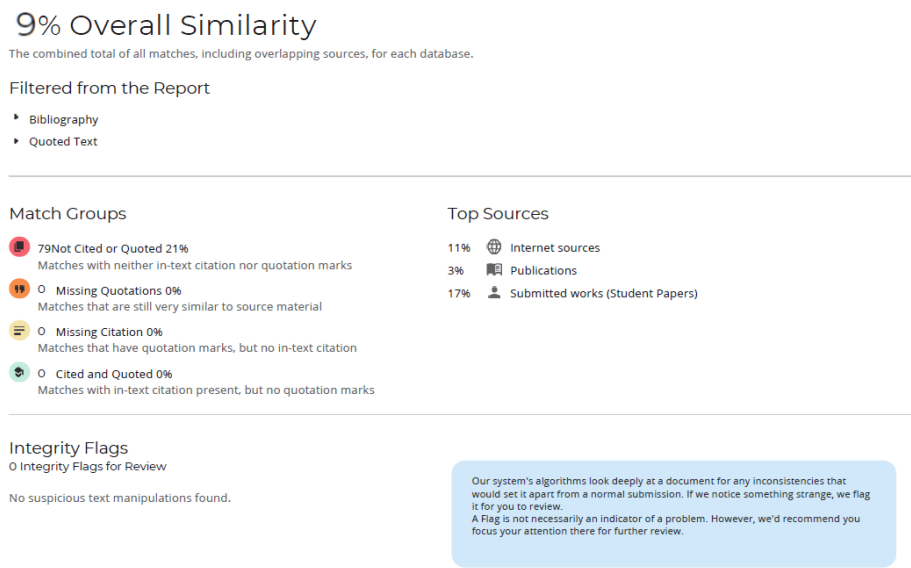


Figure 10.1: Screenshot of the Plagiarism Report

CHAPTER 11

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
OCR	Optical Character Recognition
NLP	Natural Language Processing
ML	Machine Learning
REST	REpresentational State Transfer
LLM	Large Language Models
DRF	Django REST Framework
API	Application Programming Interface
JSON	JavaScript Object Notation

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Comprehensive Survey on Optical Character Recognition (OCR)

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Abstract—The verification and documentation of real estate transactions in the finance sector is a time-consuming and susceptible to errors, traditionally involving the manual extraction of data from property documents to generate a report. The research within this paper looks into how OCR (Optical Character Recognition) is applied to automate extraction, validation, and processing of key data such as property addresses for scanned documents complying with legal requirements. Data verification and intelligent error detection mechanisms through which artificial intelligence and machine learning techniques are integrated into the system ensure a certain level of accuracy and reliability in document generation.

The OCR-based proposed solution reduces processing times significantly, minimizes human error, and enhances overall productivity in the legal verification workflow. This approach will optimize the creation of compliantly legal documents while automating data extraction and validation for the efficient and accurate processing of real estate transactions in the finance sector.

Index Terms—Binerization, Optical Character Recognition, Pattern Matching, Segmentation, Tesseract, Text Extraction

I. INTRODUCTION

OCR facilitates the conversion of text contained within images, scanned documents, or even PDFs into readable machine data from printed or handwriting. While OCR was specifically designed for the purpose of assisting those who are blind or suffering from any form of visual impairment, upon its invention, this technology quickly became usable in multiple fields, and its first extension comprises academic research. OCR assists in the process of extracting texts without wasting much time in retrieving information, indexing, and later analysis that can be helpful in the digitization and preservation of academic literature. For researchers, OCR can unlock vast amounts of printed material in order to enhance searchability and the manipulation of textual content to great advantage in literature reviews and data collection.

II. RELATED WORK

Author[1] analyses the fundamentals of Optical Character Recognition (OCR). They focus on the main principles and diverse applications, as can be used in banking, health care, or jurisprudence. The authors concentrate how OCR technology is applied to convert text into a digital format for enhanced readability and efficiency. A paper further focuses on the typical issues with precision and usability dependency on font type and quality of the documents for OCR processing. The authors will present an all-inclusive overview, explaining to the practitioners and researchers the status of OCR technology and the future potential advancements.

Author[2] performs a thorough case study of the Tesseract OCR tool in terms of architecture, functionalities, and performance analysis. It was established that the use of a grayscale image improves recognition significantly more than color images; this is mainly because of low levels of noise. They suggest several pre-processing techniques, such as adaptive thresholding, noise removal, etc., to enhance the quality of input before the process so that the input data is as optimal as possible. They compare Tesseract with other OCR tools in different font types and layout conditions, making their outcome beneficial for developers to design OCR solutions properly.

Author[3] does an in-depth analysis of OCR systems discussing some of the major issues widely encountered in recognizing complex languages like Arabic and Sindhi. They trace the development of OCR technology and identify areas where current systems fall short in achieving high accuracy with complicated scripts. The authors make the case for targeted research on recognition algorithms specific to these languages as well as noting that the much-vaunted potential of multilingual OCR could be more effectively exploited were

an ability to easily switch between scripts available. Their study points out the critical aspect for the usability of OCR technology: its capability to be applied more by a greater population of linguistic communities.

Author [4] talks about various OCR methods and concentrates on the specific problem of degradation in conversion from old books, poor scans, etc., into editable text. It describes how OCR was advanced and neural networks were trained to enhance recognition abilities to learn complex patterns present within degraded images. Character n-grams are stressed by Singh also, considering the contextual relationship for enhancing accuracy in using characters. This paper deals with these challenges and recent innovations, giving valuable insights to efforts made in developing better OCR performances with regards to problems concerning poor-quality inputs.

Author[5] discussed many OCR algorithms and techniques. Much knowledge regarding the basic components which result in effectively working OCR systems is provided by them. Their evaluation relates to a vast number of techniques, both classical and recent methods developed with machine learning approaches for achieving even higher recognition rates. Some themes regarding the improvement of overall performance relate to classification methods, preprocessing steps, and feature extraction. This work presents an invaluable overview of such algorithms for those researching or trying to apply or improve OCR technologies in various applications.

The author[6] provides an overview of the development history of OCR technology and points out the problems that would still prejudice it up to this date. They point out three critical phases of OCR: image acquisition, character recognition, and post-processing and state that each of these phases is essential for obtaining clear-cut results. Talking about the growing applications of OCR in different fields, the authors appeal for more research in countering the issues, including difference in fonts and poor quality documents. The work deals with the requirements for development that can enhance the reliability of OCR and usability in real-world environments.

Author [7] talks about the recent advances in OCR through a review of different recognition approaches and searching into the challenges pursued even after much advancement in handwriting character recognition. They represent the inefficiencies of the prevailing systems, which usually fail to interpret various writing styles rightly and emphasize the necessity of stronger algorithms that are capable of coping with such differences. The authors mention some areas of ongoing research with respect to enhancing the accuracy and robustness of OCR systems, especially for historical documents and low-quality inputs. Such work leads to the development of more potent and flexible OCR solutions as key areas for further exploration are identified.

III. CHALLENGES

Challenges in character recognition accuracy for OCR systems include the impact of noise, geometric distortions, and scanning artifacts, which can significantly disrupt the feature extraction process. These factors often lead to misclassification of characters, particularly in complex legal documents where precision is essential. Such inaccuracies not only compromise the integrity of the extracted text but also pose risks for legal interpretation and compliance, highlighting the need for robust preprocessing techniques to improve input quality before recognition.

A. Character Recognition Accuracy

Character recognition accuracy in OCR systems is compromised by noise, geometric distortions, and scanning artifacts, which disrupt feature extraction and lead to misclassification. In legal documents, such inaccuracies can result in critical errors affecting legal interpretation and compliance, necessitating advanced preprocessing techniques to enhance input quality.

B. Handwritten Text Recognition

Handwritten text recognition poses a significant challenge for OCR systems because traditional algorithms are typically designed for printed text, lacking the robustness needed to accommodate the wide variability in handwriting styles. Factors such as inconsistent letter formation, slant, and spacing can lead to high error rates in character recognition. Additionally, the contextual nuances and ligatures often present in handwritten text complicate the segmentation and classification processes. As a result, achieving accurate recognition of handwritten annotations in legal documents requires specialized algorithms and training datasets that can effectively learn from diverse handwriting examples.

C. Input Document Quality Variability

Input document quality variability is a critical challenge for OCR systems, as the performance heavily relies on the clarity and fidelity of scanned images. Lower-quality scans may exhibit issues such as blurriness, skewness, and compression artifacts, which can obscure text and disrupt the uniformity of character shapes. These artifacts hinder the efficacy of feature extraction algorithms, leading to inaccurate segmentation and classification of characters. Consequently, ensuring high-quality input documents is essential for maintaining OCR accuracy, particularly in sensitive applications like legal document processing where precision is paramount.

D. Data Privacy and Compliance

Data privacy and compliance are paramount when processing sensitive legal information through OCR systems, as they must adhere to strict regulations like GDPR and HIPAA. Any unauthorized access or data breaches can result in severe legal consequences and damage to client trust, necessitating robust security measures, such as encryption and access controls, to protect confidential information.

E. Multilingual Recognition Challenges

Multilingual recognition challenges arise in OCR due to the need for distinct models and preprocessing techniques for different languages and scripts. This complexity increases the risk of errors, as variations in character sets and writing systems can hinder accurate recognition. Effective multilingual OCR requires specialized training and adaptation to accommodate diverse linguistic features.

IV. APPLICATIONS OF OCR

Optical Character Recognition is widely applied in different fields:

A. Document Digitization and Archiving

OCR is widely used for digitizing printed documents, enabling efficient archiving, searching, and retrieval of information. It has become essential in preserving historical documents and converting academic papers, books, and legal documents into editable and searchable formats.

B. Banking and Finance

In the financial sector, OCR automates the extraction of key information from checks, invoices, receipts, and other documents, improving workflow efficiency and reducing human errors. OCR is also used for processing forms and KYC documents in banking.

C. Healthcare

OCR is used to digitize medical records, prescriptions, and billing information, facilitating quicker retrieval of patient data, reducing manual paperwork, and ensuring better patient care by minimizing errors in transcription.

D. Automated Data Entry

OCR streamlines data entry processes by extracting and processing text from forms, invoices, and applications, reducing time spent on manual data entry while improving accuracy.

E. License Plate Recognition

OCR is applied in intelligent transportation systems to recognize and process vehicle license plates for toll collection, traffic enforcement, and parking management systems.

F. Assistive Technology

OCR assists individuals with visual impairments by converting printed or handwritten text into speech or Braille, providing better accessibility to books, articles, and other textual materials.

G. Translation and Text Analysis

OCR systems are integrated with translation tools to translate text from images, such as signs, menus, or scanned documents, into multiple languages, facilitating communication in multilingual environments.

These applications highlight OCR's role in automating tasks, enhancing productivity, and improving accessibility in various sectors.

CONCLUSION

The survey provides a comprehensive examination of the advancements and enduring challenges in Optical Character Recognition (OCR) technology, particularly within the context of automated legal document generation. While significant strides have been made in improving recognition accuracy and processing efficiency, critical challenges such as character recognition fidelity, variability in input document quality, the intricacies of handwritten text recognition, data privacy compliance, and the complexities of multilingual support remain formidable barriers. Overcoming these obstacles necessitates a concerted effort toward the development of sophisticated, domain-specific models that leverage advanced machine learning techniques, enhanced preprocessing strategies, and rigorous security frameworks. By addressing these multifaceted challenges, we can significantly elevate the efficacy of OCR systems in legal applications, thereby fostering greater operational efficiency, ensuring adherence to regulatory standards, and ultimately enhancing access to legal information in a dynamic and increasingly digital environment. This holistic approach not only underscores the potential of OCR technology but also highlights the imperative for ongoing research and collaboration in this vital domain.

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